

PLANETARY SCIENCE

How young is volcanism on the Moon?

Volcanic glass beads date lunar magmatism to 120 million years ago

By Yuri Amelin¹ and Qing-Zhu Yin²

The Moon is the extraterrestrial body that is not only the closest to Earth, but also the best sampled and best explored celestial object. Analyses of rocks and soils delivered by the Apollo and Luna missions have established that the Moon is ancient, dry, and depleted of volatile elements. Studies also have indicated that early in its history, the Moon was covered with molten rock. This magma ocean eventually cooled and produced a compositionally diverse surface crust (1). How long the Moon produced magma has been an open question. On page 1077 of this issue, Wang *et al.* (2) report that volcanism on the Moon occurred as recently as 120 million years ago (Ma). This implies that the Moon may still be able to produce magma.

The time limits of lunar magmatism (the production and movement of molten rock) have been pushed at both older and younger ends of the range according to advances in dating methods and expansion of collected samples. The early crust-forming magmatism that produced highland rocks spanned about 200 million years, reaching its maximum around 4.35 billion years ago (Ga) (3). Geochronological studies showed that lunar volcanism that formed the large, visible basaltic plains (called maria) at the near side of the Moon peaked around 4.0 to 3.8 Ga and ceased by 3.1 Ga (4). However, thermal models (5) and studies of lunar maria stratigraphy (6) suggested that maria volcanism probably continued at a decreasing rate to 2 Ga. Indeed, dating lunar basaltic meteorites (lava rock from mare regions) and gabbroic meteorites (rocks crystallized from magma that cooled slowly deeper in the crust) yielded a time of formation of 2.9 to 2.8 Ga ago (7, 8). The ages of basaltic rocks delivered by the Chang'e-5 mission in 2020 from the previously unsampled location northeast of Mons Rümker in northern Oceanus Procellarum region of the Moon are still younger, produced at 2.03 to 1.96 Ga (9, 10).

Determining the time range of lunar volcanism can be expanded by studying volcanic glass beads produced by lava fountains (eruptions) and preserved in lunar regolith (surface rock and debris) (11). These beads,

despite sizes of only a few tens of micrometers, can be dated with argon ($^{40}\text{Ar}/^{39}\text{Ar}$) and uranium-lead (U-Pb) radioactive isotope decay methods (12, 13). The problem, however, is the presence of impact melt glasses in the regolith samples. Throughout its history, the Moon experienced impacts by asteroids and meteoroids that melted lunar soils and rocks. The melts then became part of lunar soil. Impact glass beads inform the history of lunar bombardment (13, 14), but their presence complicates studies of rarer volcanic glass beads.

“The effort...put into finding the volcanic ‘needle’ in the impact-generated ‘haystack’ has paid off.”

Distinguishing volcanic from impact-generated glasses is challenging. Both are solidified droplets of melt from similar rocks. Although produced by different mechanisms, their chemical and isotopic compositions are generally similar. Thus, their identification relies on more subtle differences. Unfortunately, none of these differences alone is conclusive, and therefore multiple criteria were used by Wang *et al.* to identify volcanic glasses among ~3000 glass beads, and to verify their origin. Most of the beads were determined to be the result of impact melting on the basis of mineralogical and chemical evidence of incomplete melting (characteristic of very rapid and transient impact melting) and unusually high and variable concentrations of elements such as nickel that probably originated from the impactor. Thirteen glass beads were further assessed using sulfur isotope ($^{34}\text{S}/^{32}\text{S}$) composition and its correlation with sulfur concentration. These determinations distinguished beads that arose from volcanic igneous differentiation—the chemical changes in molten rock during partial melting, cooling, and movement—and those formed from impact melting. The authors ultimately identified three volcanic glass beads that yielded consistent U-Pb ages between 135 ± 14 Ma and 116 ± 12 Ma. These ages are within the range of those for Chang'e-5 impact glass beads (14), but combined evidence from mineralogy, chemical composition, sul-

fur isotopes, and extraordinarily radiogenic Pb isotopes indicate that a volcanic origin is most likely. The use of sulfur isotopes to deduce the origin of glass beads is promising not only for the future lunar studies, but also for samples from other small celestial bodies.

The findings of Wang *et al.* indicate that the Moon was still volcanically active 120 Ma ago. Indeed, some irregular mare patches on the Moon were suggested to be 100-Ma-old volcanic features on the basis of remote-sensing data (15). Volcanic glass beads generated by lava fountains and distributed by subsequent impact transportations over wider areas on the Moon, and sampled by Chang'e-5, provided the best opportunities for the authors to verify young volcanic activities on the basis of direct measurements, without a separate, dedicated spacecraft sampling mission to a specific irregular mare patch on the Moon. Wang *et al.* noted the high potassium (K), rare earth elements, phosphorous (P), sodium (Na), and thorium (Th) concentrations in the three volcanic glass beads, and further speculated that the source was also rich in these elements. Detailed modeling is needed to fully evaluate whether the estimated heat generated within the Moon is adequate to maintain local magma activities until 120 Ma.

The effort that Wang *et al.* put into finding the volcanic “needle” in the impact-generated “haystack” has paid off. It will add to the ongoing debates about thermal evolution of planetary bodies and the question of how planets cool, which is not yet known. ■

REFERENCES AND NOTES

1. L. T. Elkins-Tanton, S. Burgess, Q.-Z. Yin, *Earth Planet. Sci. Lett.* **304**, 326 (2011).
2. B.-W. Wang *et al.*, *Science* **385**, 1077 (2024).
3. L. E. Borg, R. W. Carlson, *Annu. Rev. Earth Planet. Sci.* **51**, 25 (2023).
4. S. R. Taylor, *Lunar Science - A Post-Apollo View* (Pergamon, 1975).
5. C. K. Shearer *et al.*, *Rev. Mineral. Geochem.* **60**, 365 (2006).
6. H. Hiesinger, J. W. Head III, U. Wolf, R. Jaumann, G. Neukum, *J. Geophys. Res. Planets* **108**, 5065 (2003).
7. V. A. Fernandes, R. Burgess, G. Turner, *Meteorit. Planet. Sci.* **38**, 555 (2003).
8. L. E. Borg, C. K. Shearer, Y. Asmerom, J. J. Papike, *Nature* **432**, 209 (2004).
9. X. Che *et al.*, *Science* **374**, 887 (2021).
10. Q. L. Li *et al.*, *Nature* **600**, 54 (2021).
11. C. K. Shearer, J. J. Papike, *Geochim. Cosmochim. Acta* **57**, 4785 (1993).
12. T. S. Culler, T. A. Becker, R. A. Muller, P. R. Renne, *Science* **287**, 1785 (2000).
13. A. A. Nemchin *et al.*, *Geochim. Cosmochim. Acta* **321**, 206 (2022).
14. T. Long *et al.*, *Sci. Adv.* **8**, eabq2542 (2022).
15. S. E. Braden *et al.*, *Nat. Geosci.* **7**, 787 (2014).

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